2.6.2 From Robustness to Adaptation

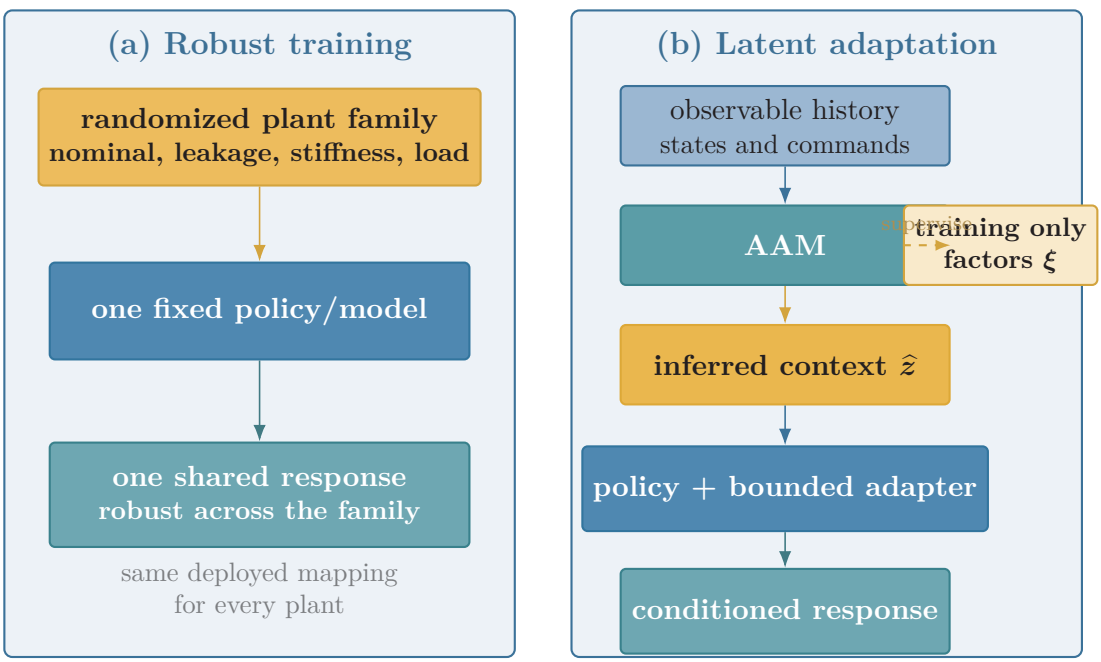


Figure 2.4: Comparison of robust training and latent adaptation. Robust training applies one fixed response across the randomized plant family. The adaptive stack uses privileged factors only to supervise training; during deployment, the AAM infers context from observable history and conditions the downstream response.

Figure 2.4 contrasts one policy trained to be robust across the plant family with an adaptive stack that conditions its response on the current plant. Teacher–student methods encode known simulation factors into a privileged latent and train an estimator to recover it from observable history. Lee et al. [Lee 20] used this construction for hidden terrain properties; Kumar et al. [Kumar 21] developed rapid motor adaptation. Unlike classical online parameter estimation [Åström 95], the learned latent need not have a direct physical interpretation. It can also condition a dynamics model rather than a policy, as in the contextual neural ODE of Yu et al. [Yu 26].

The AAM in Chapter 5 estimates a two-dimensional latent from one second of causal observable history. In AIF, that estimate enters the actor observation and a bounded adapter around frozen inverse dynamics; it does not condition the ROM, forward model or PhysX plant. The experiments keep two distinctions explicit: randomized training is robust but not automatically adaptive, and an informative privileged latent does not prove that an inferred latent improves closed-loop control.

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2.7 Reinforcement Learning and Structured Priors

2.7.1 Policy Optimization

Reinforcement learning optimizes a policy from observations to actions for expected cumulative reward [Sutton 18]. This thesis uses PPO [Schulman 17] with generalized